DeepCode AI/Graphite

AI Powered Code Review Tools

Author: Parag Agnihotri, AWS Certified Developer

mailto: parag.agnihotri@yash.com

DATE, 2025
YASH TECHNOLOGIES PVT. LTD.

Table of Contents

1 *DeepCode AI* 2

1.1 Introduction 2

1.2 Features 2

1.3 Working 3

1.4 Implementation 4

2 *Graphite* 7

2.1 Introduction 7

2.2 Features 7

2.3 Working 9

@2025 Yash Technologies Pvt. Ltd., All rights reserved

DeepCode AI

1.1 *Introduction*

DeepCode AI *is an advanced code analysis and vulnerability detection platform that has been integrated into* ***Snyk's*** *developer security platform. It offers semantic code analysis capabilities that help developers identify and fix vulnerabilities in their source code. DeepCode AI leverages machine learning and AI techniques to provide real-time security analysis during the development process, functioning as a sophisticated static application security testing (SAST) solution.*

The platform was built on research from ETH Zurich, combining sophisticated code analysis techniques with machine learning models trained on millions of code repositories. This approach enables DeepCode AI to detect a wide range of security vulnerabilities, bugs, and code quality issues with high accuracy and minimal false positives.

1.2 *Features*

- ***AI Powered Code Analysis***

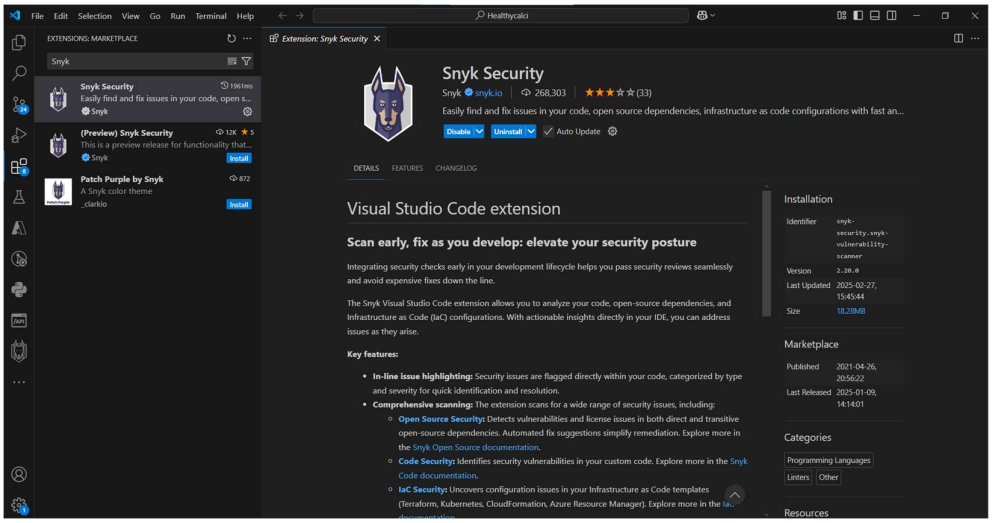
  DeepCode AI analyzes code semantically, understanding not just syntax but also the underlying meaning and execution paths. Supports multiple languages including JavaScript, TypeScript, Python, Java, PHP, and many more.

- ***Security Vulnerability Detection***

  DeepCode AI detects security vulnerabilities and authentication issues. It traces data through the application to identify how untrusted inputs might lead to security vulnerabilities.

- ***Developer Experience***

  *Integrates with popular IDEs (**VS Code**, IntelliJ, etc.) to provide real-time feedback during coding.*



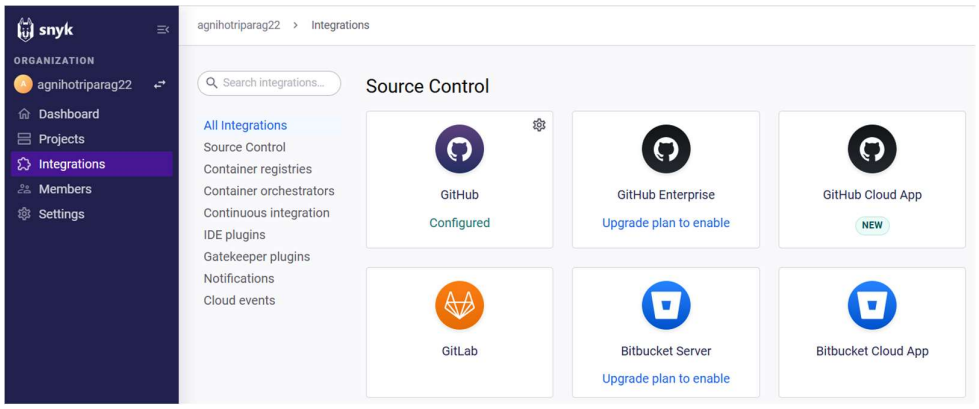
@2025 Yash Technologies Pvt. Ltd., All rights reserved

- *Quality Assurance*

  Identifies potential bugs, logic errors, and code smells. Allows organizations to define and enforce their own coding standards.

- *Integration*

  Works with GitHub, GitLab, Bitbucket, and other source code management systems. It also integrates with other tools like Jenkins, CircleCI, Travis CI, and other CI/CD platforms.

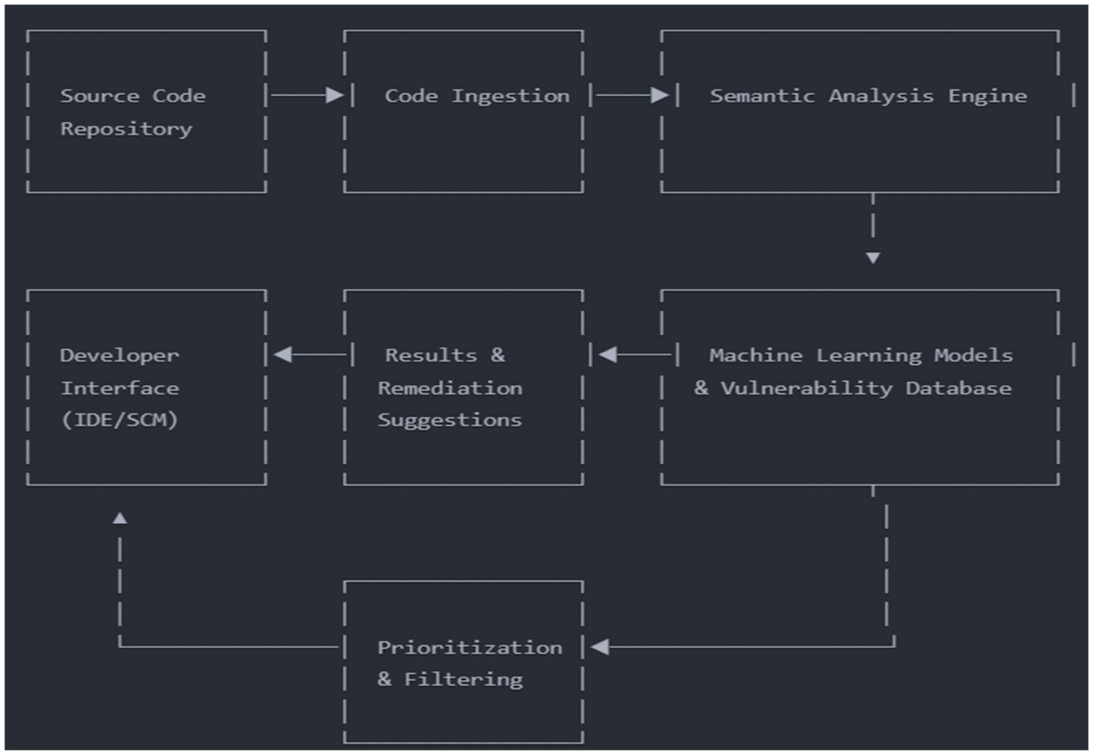


1.3 *Working*

DeepCode AI operates through a sophisticated multi-stage process that combines:

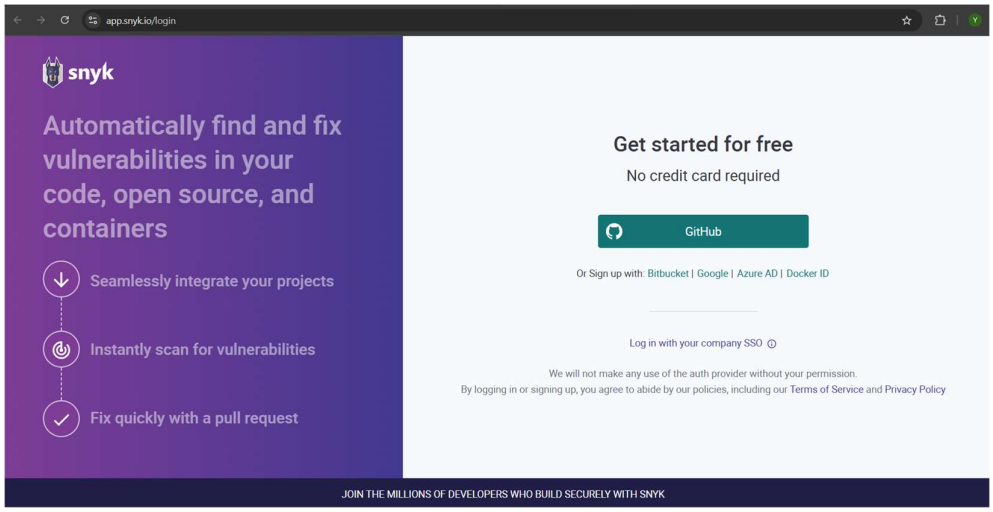
- ***Code Ingestion:*** *The system ingests source code from repositories, IDEs, or CI/CD pipelines.*
- ***Semantic Analysis:*** *DeepCode AI performs deep semantic analysis of the code, understanding Program Flow, Data Flows, Function calls, etc.*
- ***Machine Learning Analysis:*** *The code is analyzed using ML models trained on millions of code repositories to identify patterns.*
- ***Vulnerability Detection:*** *The system identifies security vulnerabilities and bugs by tracing untrusted inputs through the application and analyzing how data is processed and manipulated.*
- ***Feedback Generation:*** *DeepCode AI generates contextual feedback including detailed description of the issue, specific code location, explanation of the vulnerability, suggested remediation steps*
- ***Developer Notification:*** *Results are presented to developers through their preferred interface IDE, CLI, etc.*

@2025 Yash Technologies Pvt. Ltd., All rights reserved



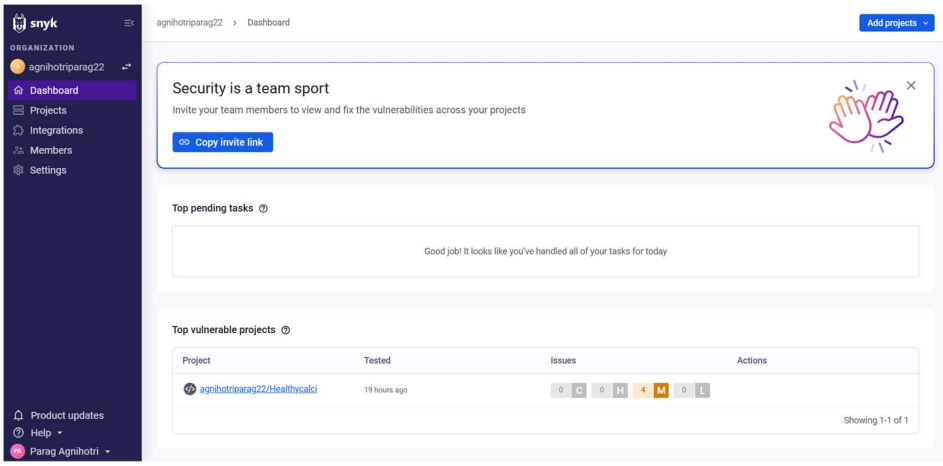
1.4 *Implementation*

- ***Signup:*** *Start by signing into Synk with GitHub, Bitbucket, Google, etc.*

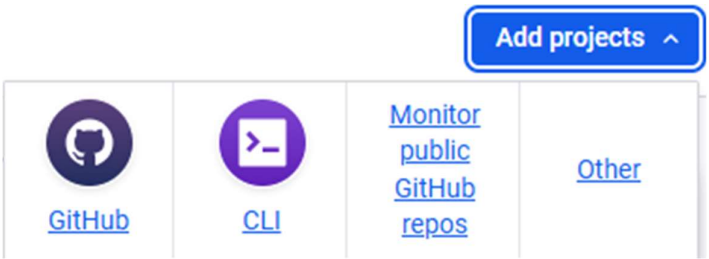


- ***Dashboard:*** *Navigating through various features of the dashboard like:*
  - *Add Projects*
  - *Integrate with various tools*
  - *Setup(Extension/CLI)*
  - *Invite Team Members, etc*

@2025 Yash Technologies Pvt. Ltd., All rights reserved

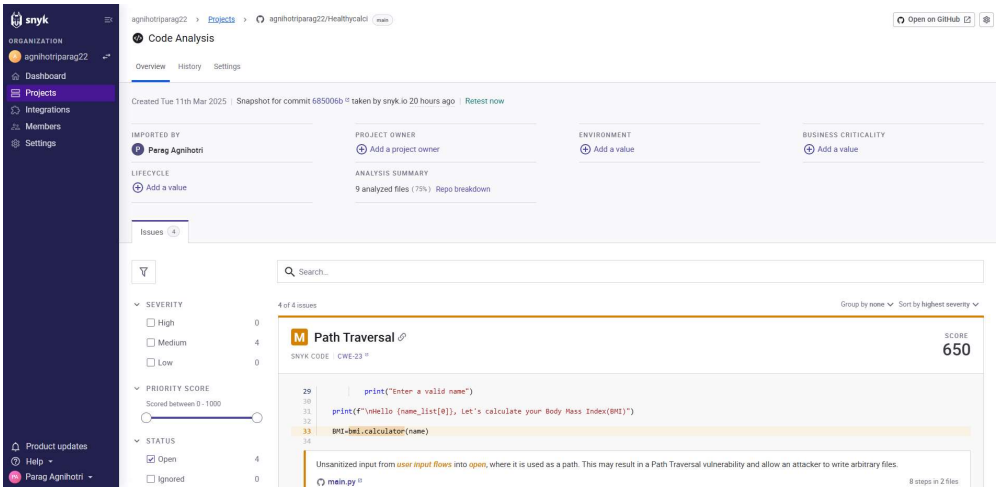


- ***Adding Members:*** *You can invite your teammates to work on a project.*
- ***Adding Projects:*** *Add projects to your workspace from GitHub or other source Providers.*



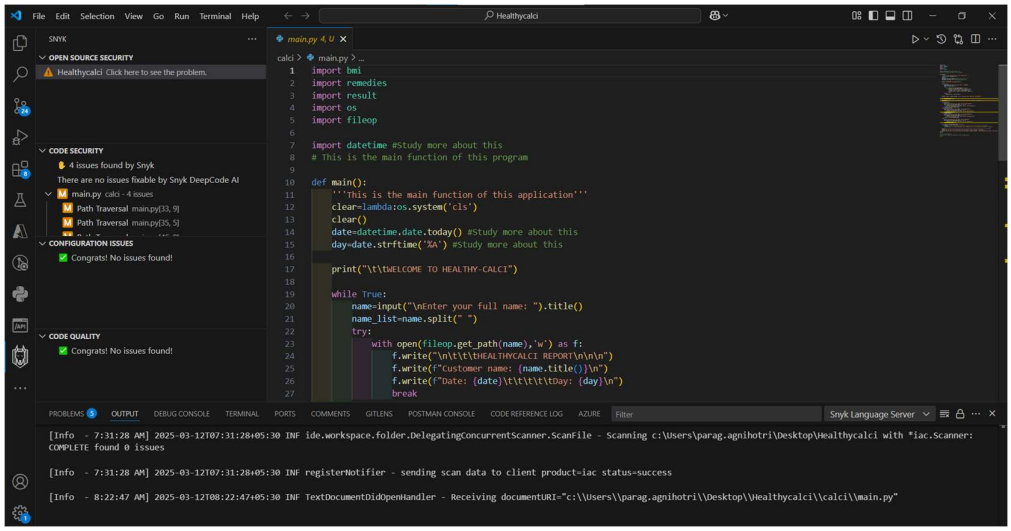
- ***Working with various projects:*** *You can work on various projects from the CLI, Extension, or Web itself.*

Snyk's UI:

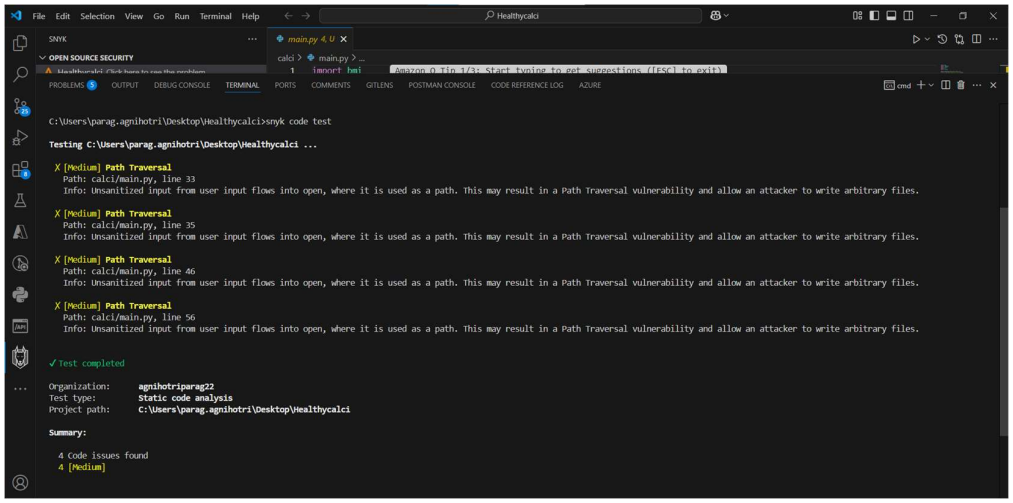


@2025 Yash Technologies Pvt. Ltd., All rights reserved

Snyk's Extension:



CLI:



Use command using npm(step by step):

1. Command to install snyk on your system:

   npm install -g snyk

2. Command to authenticate account:

   snyk auth

3. Command to test your code:

   snyk code test --org=6415a370-cfe2-47bd-ba96-c9d02e8f7224

4. Use this command to monitor your projects(Optional):

   snyk monitor --all-projects --org=6415a370-cfe2-47bd-ba96-c9d02e8f7224

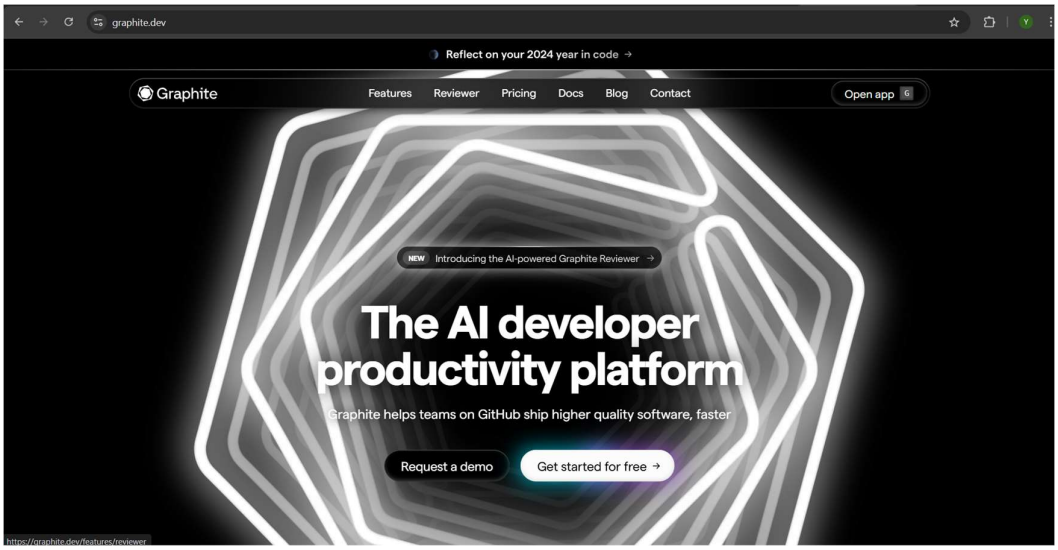
@2025 Yash Technologies Pvt. Ltd., All rights reserved

Graphite.dev

1.1 Introduction

Graphite.dev is a modern engineering platform designed to streamline the code review and pull request (PR) workflow for software development teams. Graphite offers a developer-centric approach to version control and code collaboration that integrates with existing GitHub repositories.

Graphite addresses common pain points in the traditional Git workflow, particularly around managing feature branches, code reviews, and pull requests. By providing a more intuitive interface and powerful tools specifically designed for handling complex development workflows, Graphite aims to reduce the cognitive load on developers and speed up the review process.



1.2 Features

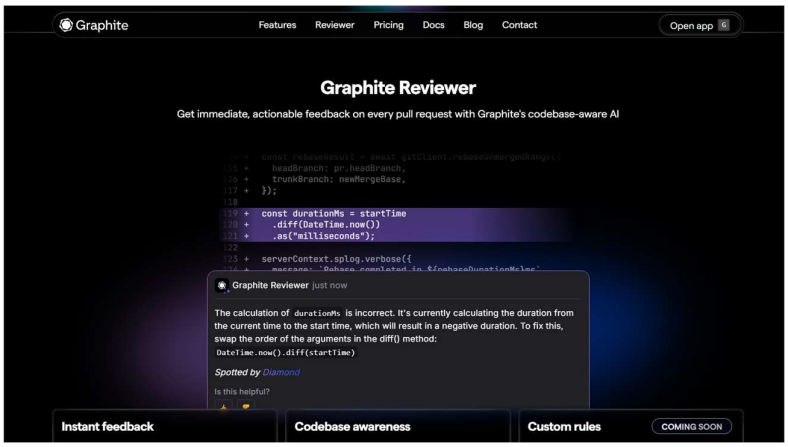
- ***Stack-Based Development***

  Allows developers to work on multiple dependent changes simultaneously without waiting for code reviews. It also manages dependencies between related changes. It Simplifies handling of multiple related branches through a visual interface.

- ***Enhanced Code Review Experience***

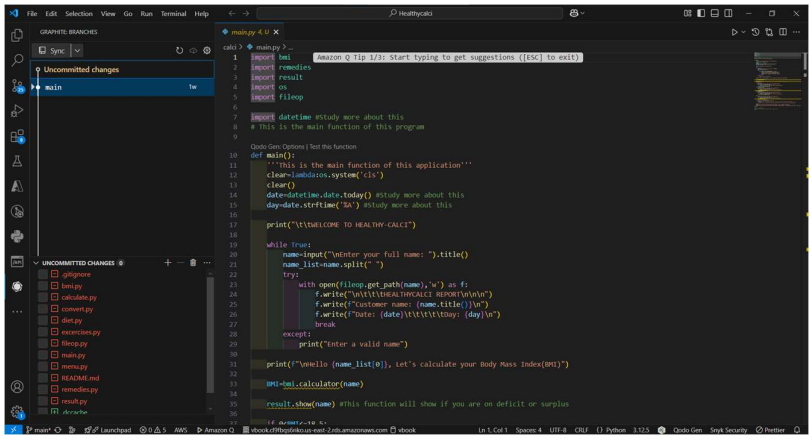
  Provides a clear, side-by-side comparison of code changes. It also keeps track of review status and feedback across multiple changes, last but not least it Alerts team members about pending reviews and updates.

@2025 Yash Technologies Pvt. Ltd., All rights reserved



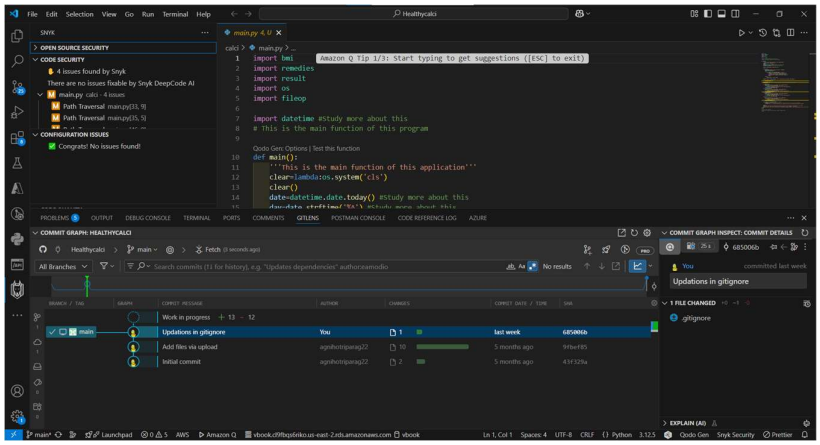
- *GitHub Integration*

  Synchronizes with GitHub repositories without disrupting existing workflows. It Works with existing CI/CD pipelines and GitHub Actions. It creates and manages GitHub pull requests directly from the Graphite interface.



- *Command Line Interface(CLI)*

  Offers a command-line interface for developers who prefer terminal-based workflows. Functions as an extension to Git, adding stack-based commands. We can also keep track of changes and Repository worktree through Gitlens.



@2025 Yash Technologies Pvt. Ltd., All rights reserved

Steps to Configure Graphite CLI:

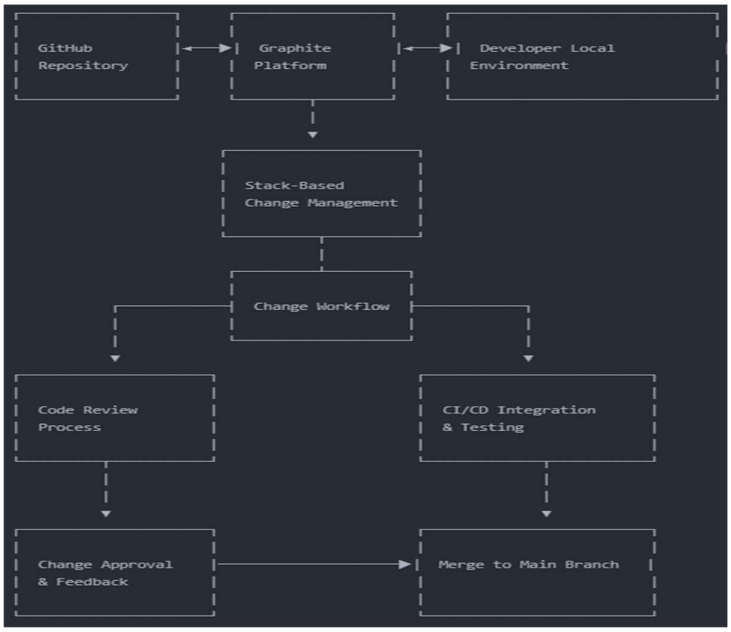
1. *Run the following command in your terminal to install Graphite in your directory.*
   npm install -g @withgraphite/graphite-cli@stable
2. *Authenticate using unique token: Create an authentication token (from github from the given create token button which will redirect you to GitHub token generation page) below and register it in your terminal using the corresponding Graphite authentication command.*

- *Team Collaboration Tools*
  Provides visibility into team activity and pending reviews. You can track review times, merge rates, and other team performance metrics.
- *Developer Productivity Features*
  Helps developers manage their in-progress work. It also Provides tools to easily resolve merge conflicts.

1.3 Working

- ***Repository Connection:*** *Developers connect their GitHub repositories to Graphite.*
- ***Stack Creation:*** *Instead of creating isolated feature branches, developers create "stacks" of related changes. (Each change in a stack is a discrete unit of work)*
- ***Change Development:*** *Developers work on changes locally.*
- ***Dependency Management:*** *Graphite automatically tracks dependencies between changes for example When a parent change is updated, child changes are rebased*
  And we can identify and resolve merge conflicts through the interface.
- ***Code Review Process:*** *Changes are submitted for review individually or as a stack and the reviewers can see the entire context of a change, comments and feedback are tracked per change as well.*
- ***GitHub Synchronization:*** *Approved changes are synchronized with GitHub, Graphite creates pull requests in GitHub and also tracks merged PRs.*
- ***Continuous Integration:*** *Changes flow back to the main branch when approved and tested.*

@2025 Yash Technologies Pvt. Ltd., All rights reserved



@2025 Yash Technologies Pvt. Ltd., All rights reserved